BrucellaID
Brucella species identification and MLST from reads

# Quality assurance report

4 sample(s): 4 ok, 0 with a warning, 0 failed. BrucellaID 2.1.0, run on 2026-11-08T10:55:07-05:00.

**This is an example report: a real run of the four public read sets of the BrucellaID tutorial, produced by scripts/make_example_report.py, which leaves out the computer it ran on. The user account, the host name, the operating system and the working directory read [removed], and the input and reference files are named without the folder they were in. Everything else is as BrucellaID wrote it. A report of your own names them all, which is what makes it worth signing.**

## 1 Summary

| | |
|---|---|
| Run started | 2026-11-08T10:55:07-05:00 |
| Run finished | 2026-11-08T10:55:47-05:00 |
| Operator | M-O Duceppe |
| Command | python3 scripts/make_example_report.py reads |

| SAMPLE | SPECIES | FROM | DECISION | ST | ALLELE PROFILE | CLOSEST REFERENCE | STATUS |
|---|---|---|---|---|---|---|---|
| **Brucella abortus · ST 1 (1 sample(s))** | | | | | | | |
| ERR3454415 | Brucella abortus | MASH+ST | agree | 1 | 2-1-1-2-1-3-1-1-1 | GCA_000742195.1_ASM74219v1_genomic.fna.gz | ok |
| **Brucella canis · ST 20 (1 sample(s))** | | | | | | | |
| ERR2136546 | Brucella canis | MASH+ST | arbitrated | 20 | 1-6-4-1-5-3-5-4-6 | GCA_000530495.1_Oliveri_genomic.fna.gz | ok |
| **Brucella melitensis · ST 11 (1 sample(s))** | | | | | | | |
| SRR19520329 | Brucella melitensis | MASH+ST | agree | 11 | 3-2-3-2-1-5-3-2-10 | GCA_061325495.1_ASM6132549v1_genomic.fna.gz | ok |
| **Brucella suis · ST 15 (1 sample(s))** | | | | | | | |
| SRR8550484 | Brucella suis | MASH+ST | agree | 15 | 1-2-7-1-3-3-5-3-2 | GCA_000742055.1_ASM74205v1_genomic.fna.gz | ok |

REVIEWED BY

Name Date Signature

## 2 Notes, warnings and errors

| SAMPLE | STATUS | WHAT IT SAYS |
|---|---|---|
| ERR2136546 | ok | nearly as close: GCA_000741995.1_ASM74199v1_genomic.fna.gz (Brucella suis, 986/1000 hashes), 3 other Brucella canis genome(s) |
| ERR2136546 | ok | MASH cannot separate Brucella canis and Brucella suis: ST 20 is Brucella canis in 70 of 70 isolate(s), which agrees with the closest reference |
| ERR3454415 | ok | nearly as close: 9 other Brucella abortus genome(s) |
| SRR19520329 | ok | nearly as close: 6 other Brucella melitensis genome(s) |
| SRR8550484 | ok | nearly as close: 2 other Brucella suis genome(s) |

# 3 Samples

## ERR3454415 ok Brucella abortus ST 1

### IDENTIFICATION: THE MASH DISTANCE

| | |
|---|---|
| Closest reference | GCA_000742195.1_ASM74219v1_genomic.fna.gz |
| What it is | Brucella abortus str. 2308 |
| NCBI accessions | GCA_000742195.1 |
| MASH distance | 0.000144 |
| Estimated identity | 99.986% |
| Shared min-hashes | 994/1000 (at least 150 required) |
| K-mer copies required | 14 (from the estimated 70x coverage) |

### THE SPECIES CALL

| | |
|---|---|
| Species reported | Brucella abortus |
| Decided by | MASH+ST |
| Species within reach | Brucella abortus |
| Sequence type | 1 |
| What that ST is | ST 1 is Brucella abortus in 463 of 466 isolate(s) (the rest: Brucella melitensis:3) |
| Biovars of its isolates | 1:86, 2:16, 3:1, 4:7 |

**agree** The sequence type names the same species as the closest reference genome, which named it on its own: the two kinds of evidence agree.

Closest first. A shaded row is too close to the first to be separated by distance alone. The other 422 reference(s) are further away; brucellaid_mash.tsv has every distance.

| REFERENCE GENOME | SPECIES | STRAIN | DISTANCE | IDENTITY % | SHARED HASHES |
|---|---|---|---|---|---|
| GCA_000742195.1_ASM74219v1_genomic.fna.gz | Brucella abortus | 2308 | 0.000144 | 99.986 | 994/1000 |
| GCA_050497595.1_ASM5049759v1_genomic.fna.gz | Brucella abortus | Ba44 | 0.000337 | 99.966 | 986/1000 |
| GCA_000739315.1_ASM73931v1_genomic.fna.gz | Brucella abortus | 544 | 0.000361 | 99.964 | 985/1000 |
| GCA_000370045.1_Bruc_abor_63_168_V1_genomic.fna.gz | Brucella abortus | 63/168 | 0.000386 | 99.961 | 984/1000 |
| GCA_000472245.1_Denovo_assembly_genomic.fna.gz | Brucella abortus | S99 | 0.000533 | 99.947 | 978/1000 |
| GCA_039649615.1_ASM3964961v1_genomic.fna.gz | Brucella abortus | X20_S4 | 0.000533 | 99.947 | 978/1000 |
| GCA_050498085.1_ASM5049808v1_genomic.fna.gz | Brucella abortus | Ba732 | 0.000533 | 99.947 | 978/1000 |
| GCA_022426865.1_ASM2242686v1_genomic.fna.gz | Brucella abortus | BANDDB_5033 | 0.000582 | 99.942 | 976/1000 |
| GCA_030761375.1_ASM3076137v1_genomic.fna.gz | Brucella abortus | LBAB008 | 0.000607 | 99.939 | 975/1000 |
| GCA_022426825.1_ASM2242682v1_genomic.fna.gz | Brucella abortus | BANDDB_3995 | 0.000631 | 99.937 | 974/1000 |

### MLST: THE SEQUENCE TYPE

| | |
|---|---|
| Sequence type | 1 |
| Allele profile | 2-1-1-2-1-3-1-1-1 |
| Every locus called | yes |
| Reads in the input | 1,175,082 |
| Reads matching an allele | 2,948 |

### INPUT FILES

| FILE | SIZE | MODIFIED |
|---|---|---|
| ERR3454415_1.fastq.gz | 87.4 MB | 2026-11-07 15:47:04 EST |
| ERR3454415_2.fastq.gz | 83.5 MB | 2026-11-07 15:48:02 EST |

```
ERR3454415_1.fastq.gz
MD5 8eaddc2c8044657712ec586ba5482c70
ERR3454415_2.fastq.gz
MD5 b2706df339382a63f38bba34222b4053
```

The margin each call had over the next allele of its locus; a ratio near 1.00 is the one to check.

| LOCUS | ALLELE | READS | RUNNER-UP (ITS READS) | AHEAD | RATIO | NOTE |
|---|---|---|---|---|---|---|
| gap | 2 | 432 | gap_6 (300) | 132 | 1.44 | |
| aroA | 1 | 322 | aroA_30 (210) | 112 | 1.53 | |
| glk | 1 | 222 | glk_21 (158) | 64 | 1.41 | |
| dnaK | 2 | 340 | dnaK_26 (222) | 118 | 1.53 | |
| gyrB | 1 | 334 | gyrB_22 (224) | 110 | 1.49 | |
| trpE | 3 | 320 | trpE_4 (242) | 78 | 1.32 | |
| cobQ | 1 | 308 | cobQ_12 (196) | 112 | 1.57 | |
| int_hyp | 1 | 368 | int_hyp_4 (256) | 112 | 1.44 | |
| omp25 | 1 | 292 | omp25_12 (186) | 106 | 1.57 | |

### NOTES

- nearly as close: 9 other Brucella abortus genome(s)

# ERR2136546 ok Brucella canis ST 20

### IDENTIFICATION: THE MASH DISTANCE

| | |
|---|---|
| Closest reference | GCA_000530495.1_Oliveri_genomic.fna.gz |
| What it is | Brucella canis str. Oliveri |
| NCBI accessions | GCA_000530495.1 |
| MASH distance | 0.000119 |
| Estimated identity | 99.988% |
| Shared min-hashes | 995/1000 (at least 150 required) |
| K-mer copies required | 9 (from the estimated 47x coverage) |

### THE SPECIES CALL

| | |
|---|---|
| Species reported | Brucella canis |
| Decided by | MASH+ST |
| Species within reach | Brucella canis, Brucella suis |
| Sequence type | 20 |
| What that ST is | ST 20 is Brucella canis in 70 of 70 isolate(s) |

**arbitrated** A reference of a second species was near enough to the closest one that the distance did not really choose. The sequence type chose between the species it left within reach.

Closest first. A shaded row is too close to the first to be separated by distance alone. The other 427 reference(s) are further away; brucellaid_mash.tsv has every distance.

| REFERENCE GENOME | SPECIES | STRAIN | DISTANCE | IDENTITY % | SHARED HASHES |
|---|---|---|---|---|---|
| GCA_000530495.1_Oliveri_genomic.fna.gz | Brucella canis | Oliveri | 0.000119 | 99.988 | 995/1000 |
| GCA_000741995.1_ASM74199v1_genomic.fna.gz | Brucella suis | BSQ | 0.000337 | 99.966 | 986/1000 |
| GCA_056582615.1_ASM5658261v1_genomic.fna.gz | Brucella canis | M-24 | 0.000361 | 99.964 | 985/1000 |
| GCA_000367265.1_Bruc_cani_CNGB_1172_V1_genomic.fna.gz | Brucella canis | CNGB 1172 | 0.000483 | 99.952 | 980/1000 |
| GCA_050496145.1_ASM5049614v1_genomic.fna.gz | Brucella canis | Bc1066 | 0.000483 | 99.952 | 980/1000 |

### MLST: THE SEQUENCE TYPE

| | |
|---|---|
| Sequence type | 20 |
| Allele profile | 1-6-4-1-5-3-5-4-6 |
| Every locus called | yes |
| Reads in the input | 625,576 |
| Reads matching an allele | 1,824 |

### INPUT FILES

| FILE | SIZE | MODIFIED |
|---|---|---|
| ERR2136546_1.fastq.gz | 57.0 MB | 2026-11-07 15:50:28 EST |
| ERR2136546_2.fastq.gz | 63.9 MB | 2026-11-07 15:51:19 EST |

```
ERR2136546_1.fastq.gz
MD5 7573e407362c1458a464a4d92fec2db4
ERR2136546_2.fastq.gz
MD5 a83a9dceaee29371219b98f0a2f1dc39
```

The margin each call had over the next allele of its locus; a ratio near 1.00 is the one to check.

| LOCUS | ALLELE | READS | RUNNER-UP (ITS READS) | AHEAD | RATIO | NOTE |
|---|---|---|---|---|---|---|
| gap | 1 | 232 | gap_26 (170) | 62 | 1.36 | |
| aroA | 6 | 194 | aroA_31 (80) | 114 | 2.42 | |
| glk | 4 | 148 | glk_2 (94) | 54 | 1.57 | |
| dnaK | 1 | 244 | dnaK_2 (162) | 82 | 1.51 | |
| gyrB | 5 | 224 | gyrB_4 (110) | 114 | 2.04 | |
| trpE | 3 | 178 | trpE_5 (114) | 64 | 1.56 | |
| cobQ | 5 | 172 | cobQ_30 (108) | 64 | 1.59 | |
| int_hyp | 4 | 234 | int_hyp_1 (180) | 54 | 1.30 | |
| omp25 | 6 | 172 | omp25_2 (98) | 74 | 1.76 | |

### NOTES

- nearly as close: GCA_000741995.1_ASM74199v1_genomic.fna.gz (Brucella suis, 986/1000 hashes), 3 other Brucella canis genome(s)
- MASH cannot separate Brucella canis and Brucella suis: ST 20 is Brucella canis in 70 of 70 isolate(s), which agrees with the closest reference

## SRR19520329 ok Brucella melitensis ST 11

### IDENTIFICATION: THE MASH DISTANCE

| | |
|---|---|
| Closest reference | GCA_061325495.1_ASM6132549v1_genomic.fna.gz |
| What it is | Brucella melitensis str. 48_S36 |
| NCBI accessions | GCA_061325495.1 |
| MASH distance | 0.000119 |
| Estimated identity | 99.988% |
| Shared min-hashes | 995/1000 (at least 150 required) |
| K-mer copies required | 11 (from the estimated 53x coverage) |

### THE SPECIES CALL

| | |
|---|---|
| Species reported | Brucella melitensis |
| Decided by | MASH+ST |
| Species within reach | Brucella melitensis |
| Sequence type | 11 |
| What that ST is | ST 11 is Brucella melitensis in 546 of 546 isolate(s) |
| Biovars of its isolates | 1:1, 2:7, 3:12 |

**agree** The sequence type names the same species as the closest reference genome, which named it on its own: the two kinds of evidence agree.

Closest first. A shaded row is too close to the first to be separated by distance alone. The other 425 reference(s) are further away; brucellaid_mash.tsv has every distance.

| REFERENCE GENOME | SPECIES | STRAIN | DISTANCE | IDENTITY % | SHARED HASHES |
|---|---|---|---|---|---|
| GCA_061325495.1_ASM6132549v1_genomic.fna.gz | Brucella melitensis | 48_S36 | 0.000119 | 99.988 | 995/1000 |
| GCA_002191375.1_ASM219137v1_genomic.fna.gz | Brucella melitensis | BwIM_ITA_55 | 0.000264 | 99.974 | 989/1000 |
| GCA_006517305.1_ASM651730v1_genomic.fna.gz | Brucella melitensis | 2011.TE.11844.1.1 | 0.000337 | 99.966 | 986/1000 |
| GCA_047040135.1_M2206_genomic.fna.gz | Brucella melitensis | M2206 | 0.000361 | 99.964 | 985/1000 |
| GCA_006506815.1_ASM650681v1_genomic.fna.gz | Brucella melitensis | 2011.TE.6837.1.1 | 0.000410 | 99.959 | 983/1000 |
| GCA_006506735.1_ASM650673v1_genomic.fna.gz | Brucella melitensis | 2013.TE.15003.1.1 | 0.000483 | 99.952 | 980/1000 |
| GCA_017177365.1_ASM1717736v1_genomic.fna.gz | Brucella melitensis | 759 | 0.000557 | 99.944 | 977/1000 |

### MLST: THE SEQUENCE TYPE

| | |
|---|---|
| Sequence type | 11 |
| Allele profile | 3-2-3-2-1-5-3-2-10 |
| Every locus called | yes |
| Reads in the input | 1,136,924 |
| Reads matching an allele | 2,824 |

### INPUT FILES

| FILE | SIZE | MODIFIED |
|---|---|---|
| SRR19520329_1.fastq.gz | 35.9 MB | 2026-11-07 15:45:34 EST |
| SRR19520329_2.fastq.gz | 40.6 MB | 2026-11-07 15:46:03 EST |

```
SRR19520329_1.fastq.gz
MD5 12edf200445bafd03d033fac3d98118c
SRR19520329_2.fastq.gz
MD5 a81d3abf43d7a2849c9062061cd8603a
```

The margin each call had over the next allele of its locus; a ratio near 1.00 is the one to check.

| LOCUS | ALLELE | READS | RUNNER-UP (ITS READS) | AHEAD | RATIO | NOTE |
|---|---|---|---|---|---|---|
| gap | 3 | 350 | gap_34 (250) | 100 | 1.40 | |
| aroA | 2 | 324 | aroA_1 (256) | 68 | 1.27 | |
| glk | 3 | 282 | glk_26 (184) | 98 | 1.53 | |
| dnaK | 2 | 280 | dnaK_6 (204) | 76 | 1.37 | |
| gyrB | 1 | 374 | gyrB_2 (266) | 108 | 1.41 | |
| trpE | 5 | 348 | trpE_31 (242) | 106 | 1.44 | |
| cobQ | 3 | 260 | cobQ_2 (194) | 66 | 1.34 | |
| int_hyp | 2 | 328 | int_hyp_20 (222) | 106 | 1.48 | |
| omp25 | 10 | 270 | omp25_8 (182) | 88 | 1.48 | |

### NOTES

- nearly as close: 6 other Brucella melitensis genome(s)

## SRR8550484 ok Brucella suis ST 15

### IDENTIFICATION: THE MASH DISTANCE

| | |
|---|---|
| Closest reference | GCA_000742055.1_ASM74205v1_genomic.fna.gz |
| What it is | Brucella suis str. Thompsen |
| NCBI accessions | GCA_000742055.1 |
| MASH distance | 0.000264 |
| Estimated identity | 99.974% |
| Shared min-hashes | 989/1000 (at least 150 required) |
| K-mer copies required | 14 (from the estimated 69x coverage) |

### THE SPECIES CALL

| | |
|---|---|
| Species reported | Brucella suis |
| Decided by | MASH+ST |
| Species within reach | Brucella suis |
| Sequence type | 15 |
| What that ST is | ST 15 is Brucella suis in 99 of 103 isolate(s) (the rest: Brucella melitensis:2, Brucella ovis:1, Brucella sp.:1) |
| Biovars of its isolates | 2:60 |

**agree** The sequence type names the same species as the closest reference genome, which named it on its own: the two kinds of evidence agree.

Closest first. A shaded row is too close to the first to be separated by distance alone. The other 427 reference(s) are further away; brucellaid_mash.tsv has every distance.

| REFERENCE GENOME | SPECIES | STRAIN | DISTANCE | IDENTITY % | SHARED HASHES |
|---|---|---|---|---|---|
| GCA_000742055.1_ASM74205v1_genomic.fna.gz | Brucella suis | Thompsen | 0.000264 | 99.974 | 989/1000 |
| GCA_000365565.1_Bruc_suis_F8_06-2_V1_genomic.fna.gz | Brucella suis | F8/06-2 | 0.000459 | 99.954 | 981/1000 |
| GCA_014884445.1_ASM1488444v1_genomic.fna.gz | Brucella suis | CVI_105 | 0.000582 | 99.942 | 976/1000 |
| GCA_000292105.2_ASM29210v2_genomic.fna.gz | Brucella suis | BCB032 | 0.001839 | 99.816 | 927/1000 |
| GCA_055657665.1_ASM5565766v1_genomic.fna.gz | Brucella suis | 25Ref0017 | 0.001839 | 99.816 | 927/1000 |

### MLST: THE SEQUENCE TYPE

| | |
|---|---|
| Sequence type | 15 |
| Allele profile | 1-2-7-1-3-3-5-3-2 |
| Every locus called | yes |
| Reads in the input | 1,693,672 |
| Reads matching an allele | 3,204 |

### INPUT FILES

| FILE | SIZE | MODIFIED |
|---|---|---|
| SRR8550484_1.fastq.gz | 71.6 MB | 2026-11-07 15:48:53 EST |
| SRR8550484_2.fastq.gz | 72.4 MB | 2026-11-07 15:49:45 EST |

```
SRR8550484_1.fastq.gz
MD5 68833febebe273fadd9dc6ee0e95ed16
SRR8550484_2.fastq.gz
MD5 9b8438792c71722cd7fb4fb0d18e414c
```

The margin each call had over the next allele of its locus; a ratio near 1.00 is the one to check.

| LOCUS | ALLELE | READS | RUNNER-UP (ITS READS) | AHEAD | RATIO | NOTE |
|---|---|---|---|---|---|---|
| gap | 1 | 508 | gap_14 (412) | 96 | 1.23 | |
| aroA | 2 | 284 | aroA_16 (232) | 52 | 1.22 | |
| glk | 7 | 282 | glk_4 (202) | 80 | 1.40 | |
| dnaK | 1 | 302 | dnaK_27 (224) | 78 | 1.35 | |
| gyrB | 3 | 384 | gyrB_1 (290) | 94 | 1.32 | |
| trpE | 3 | 346 | trpE_5 (296) | 50 | 1.17 | |
| cobQ | 5 | 294 | cobQ_4 (220) | 74 | 1.34 | |
| int_hyp | 3 | 444 | int_hyp_1 (272) | 172 | 1.63 | |
| omp25 | 2 | 330 | omp25_21 (250) | 80 | 1.32 | |

### NOTES

- nearly as close: 2 other Brucella suis genome(s)

## 4 Definitions and methods

### The references listed for each sample

A sample block lists the five closest reference genomes, and any other near enough to the closest that the distance does not separate them, which is the evidence the species call rests on. brucellaid_mash.tsv holds the distance to every reference of the sketch.

### The species, from a MASH distance

The reads of a sample are reduced to the 1000 smallest hashes of their 21-mers, ignoring the k-mers seen fewer times than the copy threshold derived from the coverage, and that sketch is compared with the sketch of each reference genome. The distance is an estimate of the fraction of positions that differ, so 0.000024 is about 99.998% identity. The closest reference names the species, provided the reads share at least the required number of its 1000 min-hashes; below that the sample is reported as ND, because the closest genome of the set is not an answer. The reference data table above says which set this run measured against.

### Why there is no biovar in this report

BrucellaID does not report a biovar, because it cannot measure one. Three things stand in the way. Hardly any reference carries a biovar at all: 7 of the 432 genomes of the default set do, because the NCBI records of the rest carry none, so 93% of the biovar-labelled B. abortus isolates of the PubMLST isolate database are of a biovar no reference has. The distance cannot separate the biovars that are there: the two B. suis biovar 2 genomes are 0.00163 apart, further than the biovar 3 genome is from one of them (0.00150). And a biovar is a phenotype that genome clustering does not track -- the best any rule from sequence type to biovar could manage over the isolate database is 66% for B. abortus and 63% for B. melitensis, because one ST spans several biovars. The biovars recorded for the ST are shown above for what they are worth. For a biovar, use classical biotyping, or brucetyper, which reads the Bruce-ladder and Suis-ladder assays in silico.

### The sequence type, from the alleles

The same reads are given to Seal with the alleles of the nine loci as references, counting each read for every allele it shares the highest number of 25-mers with and for no other. The allele the sample carries is therefore counted for every read of its locus, and a neighbouring allele only for the reads that fall where the two agree, so the allele with the most reads wins. No mismatch is allowed in a k-mer, because the alleles of a locus differ by as little as one base. Two alleles with the same count are reported as a tie rather than separated by a coin toss, and a locus below the minimum read count is not called.

### What the ST values mean

A number is a profile that is in the table. "new" means all nine alleles were called but the combination is not in the table, which is either a genuinely new ST or a novel allele called as its nearest neighbour. "ND" means at least one locus had no call; the note says which.

### How the species is decided when two are equally close

Some Brucella species are closer to each other than the variation within either of them: in the sketches that ship with BrucellaID, the closest pair of different species is always a B. canis and a B. suis: 0.00024 apart in the 16 genomes of vSNP, 0.000264 in the 432 of the default set. Where a reference of a second species is within the close-match margin of the closest one, the distance has not really chosen, and the sequence type chooses among those species, because the two share no ST. An ST is only used when enough of its isolates agree on a species, and an ST that contradicts a species the distance picked cleanly is reported as a conflict and not obeyed.

### What this report does not establish

That the sample is what the label on the tube says, that the library is not a mixture of two isolates, or that the reference data is current. It establishes what the software was given, what it measured, and what it concluded from those measurements.

# 5 Run information

| | |
|---|---|
| Operator | M-O Duceppe |
| User account | [removed] |
| Computer | [removed] |
| Operating system | [removed] |
| Working directory | [removed] |
| Started | 2026-11-08T10:55:07-05:00 |
| Finished | 2026-11-08T10:55:47-05:00 |
| Command | python3 scripts/make_example_report.py reads |

## Software

| PROGRAM | VERSION |
|---|---|
| brucellaid | 2.1.0 |
| Python | 3.12.16 |
| MASH | 2.3 |
| BBTools | 40.05 |
| Java | openjdk version "25.0.4.1" 2026-08-18 |

## Parameters

| PARAMETER | VALUE |
|---|---|
| Threads | 8 |
| Parallel samples | 2 |
| MLST | True |
| Input checksums | MD5 |

## Reference data

| DATA | WHAT IT HOLDS | SOURCE | MD5 |
|---|---|---|---|
| MASH sketch | 432 reference genome(s), k = 21, 1000 hashes | the package's | 800b94a5f74a266e7c08309e78c891a2 |
| Reference table | 432 genome(s) with their species, biovar, strain and accessions | the package's | 621f43f317759e9fb04eece7f9ec4acb |
| MLST alleles | 309 allele(s) of the 9 loci | the package's | 95fcf5e43f4cbf7ebe840c8e54bfbf46 |
| MLST profiles | 122 sequence type(s) | the package's | dd4bc06867bd58ce2a8f362755aa79a7 |
| ST to species | 118 sequence type(s) with a species | the package's | 3a6751b8a1ab1273f3bc0e19e08efb7f |

```
.../brucellaid/data/references_extended.msh
.../brucellaid/data/references_extended.tsv
.../brucellaid/data/mlst_alleles.fasta
.../brucellaid/data/mlst_profiles.tsv
.../brucellaid/data/mlst_st_species.tsv
```

## References


1. Whatmore AM, Perrett LL, MacMillan AP. Characterisation of the genetic diversity of Brucella by multilocus sequencing. BMC Microbiology 2007;7:34. doi:10.1186/1471-2180-7-34
2. Ondov BD, Treangen TJ, Melsted P, et al. Mash: fast genome and metagenome distance estimation using MinHash. Genome Biology 2016;17:132. doi:10.1186/s13059-016-0997-x
3. Jolley KA, Bray JE, Maiden MCJ. Open-access bacterial population genomics: BIGSdb software, the PubMLST.org website and their applications. Wellcome Open Research 2018;3:124. doi:10.12688/wellcomeopenres.14826.1
4. Bushnell B. BBTools. https://sourceforge.net/projects/bbmap/